

5.2 #5 Show that the Fourier sine series on $(0, l)$ can be derived from the full Fourier series on $(-l, l)$.
 Proof: By the hint, given $\phi(x)$ continuous fcn on $(0, l)$.

We may consider its odd extension

$$\tilde{\phi}(x) \stackrel{\text{def}}{=} \begin{cases} \phi(x) & x \in (0, l) \\ -\phi(-x) & x \in (-l, 0] \end{cases}$$

The full series for $\tilde{\phi}(x)$ on $(-l, l)$ is:

$$\tilde{\phi}(x) = \frac{A_0}{2} + \sum_{n=1}^{\infty} \left(A_n \cdot \cos\left(\frac{n\pi x}{l}\right) + B_n \cdot \sin\left(\frac{n\pi x}{l}\right) \right)$$

where $A_n = \frac{1}{l} \int_{-l}^l \tilde{\phi}(x) \cdot \cos\left(\frac{n\pi x}{l}\right) dx \quad n=0, 1, \dots$

$$B_n = \frac{1}{l} \int_{-l}^l \tilde{\phi}(x) \cdot \sin\left(\frac{n\pi x}{l}\right) dx \quad n=1, 2, \dots$$

By 5.2 #4 (a), (i.e. using the symmetry of $\tilde{\phi}$ and $\cos(\frac{n\pi x}{l})$),

$$A_n = 0 \quad n=0, 1, 2, \dots$$

Hence $\tilde{\phi}(x) = \sum_{n=1}^{\infty} B_n \cdot \sin\left(\frac{n\pi x}{l}\right). \quad (*)$

Now for any $0 < x < l$, $(*)$ becomes

$$\phi(x) = \sum_{n=1}^{\infty} B_n \cdot \sin\left(\frac{n\pi x}{l}\right)$$

$$= \sum_{n=1}^{\infty} \left(\frac{1}{l} \int_{-l}^l \tilde{\phi}(x) \cdot \sin\left(\frac{n\pi x}{l}\right) dx \right) \cdot \sin\left(\frac{n\pi x}{l}\right)$$

using symmetry
of $\tilde{\phi}$ and $\sin(\frac{n\pi x}{l})$

$$= \sum_{n=1}^{\infty} \left(\frac{2}{l} \int_0^l \phi(x) \cdot \sin\left(\frac{n\pi x}{l}\right) dx \right) \cdot \sin\left(\frac{n\pi x}{l}\right).$$

which is the desired.

5.2 # 8 (a) If ϕ is even, then $\phi(x) = \phi(-x)$
Differentiate with respect to x , then $\phi'(x) = -\phi'(-x)$,
i.e. $\phi'(x)$ is odd.

If ψ is odd, then $\psi(x) = -\psi(-x)$.

Diff w.r.t. x . $\psi'(x) = \psi'(-x)$ i.e. $\psi'(x)$ is even

(b) If f is even, then $f(x) = f(-x)$.

We will show $F(x) \stackrel{\text{def}}{=} \int_0^x f(s) ds$ is odd.

To establish this, we compute

$$\begin{aligned} F(x) + F(-x) &= \int_0^x f(s) ds + \int_0^{-x} f(s) ds \\ &= \int_0^x f(s) ds - \int_0^x f(s) ds = 0 \end{aligned}$$

$t = -s$

$$\Rightarrow F \text{ is odd.}$$

If g is odd, then $g(x) + g(-x) = 0$. (*)

We will show $G(x) \stackrel{\text{def}}{=} \int_0^x g(s) ds$ is even.

To establish this, we compute

$$\begin{aligned} G(-x) &= \int_0^{-x} g(s) ds \stackrel{t=-s}{=} - \int_0^x g(-t) dt \\ &= - \int_0^x g(-s) ds \\ &\stackrel{\text{by } (*)}{=} - \int_0^x (-g(s)) ds = G(x) \end{aligned}$$
$$\Rightarrow G \text{ is even.}$$

5.2 #11 Find the full Fourier series of e^x on $(-l, l)$ in its real and complex forms.

Complex form:

$$C_n = \frac{1}{2l} \int_{-l}^l e^x \cdot e^{-\frac{i\pi nx}{l}} dx$$

$$= \frac{1}{2l} \int_{-l}^l e^{(1 - \frac{i\pi n}{l})x} dx$$

$$= \frac{1}{2l} \left[\frac{1}{1 - \frac{i\pi n}{l}} e^{(1 - \frac{i\pi n}{l})x} \right]_{-l}^l$$

$$= \frac{1}{2l} \frac{1}{1 - \frac{i\pi n}{l}} (e^{l - i\pi n} - e^{-l + i\pi n})$$

$$= \frac{1}{2(l - i\pi n)} (e^l - e^{-l}) \cdot (l)^n$$

Fourier series: $e^x = \sum_{n=-\infty}^{\infty} C_n e^{\frac{i n \pi x}{l}}$

$$= \sum_{n=-\infty}^{\infty} \frac{e^l - e^{-l}}{2(l - i\pi n)} \cdot (l)^n \cdot e^{\frac{i n \pi x}{l}}$$

Real form:

(You can also simply expand the complex form.)

$$e^x = \frac{A_0}{2} + \sum_{n=1}^{\infty} \left(A_n \cdot \cos\left(\frac{n\pi x}{l}\right) + B_n \cdot \sin\left(\frac{n\pi x}{l}\right) \right)$$

$$\text{where } A_n = \frac{1}{l} \int_{-l}^l e^x \cdot \cos\left(\frac{n\pi x}{l}\right) dx \quad n=0, 1, \dots$$

$$B_n = \frac{1}{l} \int_{-l}^l e^x \cdot \sin\left(\frac{n\pi x}{l}\right) dx \quad n=1, 2, \dots$$

$$A_0 = \frac{1}{l} \int_{-l}^l e^x dx = \frac{1}{l} (e^x \big|_{-l}^l) = \frac{e^l - e^{-l}}{l}$$

If $n \geq 1$, then

$$A_n = \frac{1}{l} \int_{-l}^l e^x \cdot \cos\left(\frac{n\pi x}{l}\right) dx \stackrel{\text{def}}{=} I$$

$$= \frac{1}{l} \int_{-l}^l \cos\left(\frac{n\pi x}{l}\right) d(e^x)$$

$$= \frac{1}{l} \left[\cos\left(\frac{n\pi x}{l}\right) \cdot e^x \big|_{-l}^l - \int_{-l}^l e^x \cdot \frac{n\pi}{l} \cdot (-\sin\left(\frac{n\pi x}{l}\right)) dx \right]$$

$$= \frac{1}{l} \left[(-1)^n (e^l - e^{-l}) + \frac{n\pi}{l} \int_{-l}^l \sin\left(\frac{n\pi x}{l}\right) d(e^x) \right]$$

$$= \frac{1}{l} \left[(-1)^n (e^l - e^{-l}) + \frac{n\pi}{l^2} \left[\cancel{\sin\left(\frac{n\pi x}{l}\right)} \cdot e^x \big|_{-l}^l - \int_{-l}^l e^x \cdot \frac{n\pi}{l} \cos\left(\frac{n\pi x}{l}\right) dx \right] \right]$$

$$\text{Thus } I = \frac{1}{l} \left[(-1)^n (e^l - e^{-l}) - \frac{n^2 \pi^2}{l^2} \cdot I \right]$$

$$\Rightarrow I = \frac{(-1)^n (e^l - e^{-l}) \cdot \frac{1}{l}}{1 + \frac{n^2 \pi^2}{l^2}} = \frac{(-1)^n (e^l - e^{-l}) \cdot l}{l^2 + n^2 \pi^2}$$

$$B_n = \frac{1}{l} \int_{-l}^l e^x \cdot \sin\left(\frac{n\pi x}{l}\right) dx \stackrel{\text{def}}{=} J$$

$$= \frac{1}{l} \int_{-l}^l \sin\left(\frac{n\pi x}{l}\right) d(e^x)$$

$$= \frac{1}{l} \left[\sin\left(\frac{n\pi x}{l}\right) \cdot e^x \Big|_{-l}^l - \int_{-l}^l e^x \cdot \frac{n\pi}{l} \cos\left(\frac{n\pi x}{l}\right) dx \right]$$

$$= -\frac{n\pi}{l^2} \int_{-l}^l \cos\left(\frac{n\pi x}{l}\right) d(e^x)$$

$$= -\frac{n\pi}{l^2} \left[\cos\left(\frac{n\pi x}{l}\right) \cdot e^x \Big|_{-l}^l - \int_{-l}^l e^x \cdot -\sin\left(\frac{n\pi x}{l}\right) \cdot \frac{n\pi}{l} dx \right]$$

$$= -\frac{n\pi}{l^2} \left[(-1)^n (e^l - e^{-l}) + \frac{n\pi}{l} \int_{-l}^l e^x \cdot \sin\left(\frac{n\pi x}{l}\right) dx \right]$$

$$\text{Thus } J = -\frac{n\pi}{l^2} \cdot (-1)^n (e^l - e^{-l}) - \frac{n^2\pi^2}{l^2} J$$

$$\Rightarrow J = \frac{-\frac{n\pi}{l^2} \cdot (-1)^n (e^l - e^{-l})}{1 + \frac{n^2\pi^2}{l^2}} = \frac{(-1)^n (e^l - e^{-l}) (-n\pi)}{l^2 + n^2\pi^2}$$

Thus

$$e^x = \frac{A_0}{2} + \sum_{n=1}^{\infty} \left(A_n \cdot \cos\left(\frac{n\pi x}{l}\right) + B_n \cdot \sin\left(\frac{n\pi x}{l}\right) \right)$$

$$= \frac{e^l - e^{-l}}{2l} + \sum_{n=1}^{\infty} \left[\frac{(-1)^n (e^l - e^{-l}) \cdot l}{l^2 + n^2\pi^2} \cdot \cos\left(\frac{n\pi x}{l}\right) - \frac{(-1)^n (e^l - e^{-l}) \cdot n\pi}{l^2 + n^2\pi^2} \sin\left(\frac{n\pi x}{l}\right) \right]$$

5.4 #2 Consider the series $\sum_{n=1}^{\infty} f_n(x)$
on the finite interval, say $[a, b]$,
and assume it converges uniformly.

For simplicity I denote the partial sum
by $S_m(x) \stackrel{\text{def}}{=} \sum_{n=1}^m f_n(x) \quad m=1, 2, \dots$

I denote the infinite series by $S(x) \stackrel{\text{def}}{=} \sum_{n=1}^{\infty} f_n(x)$

(1) If $S_m(x) \rightarrow S(x)$ uniformly,
then show $S_m(x) \rightarrow S(x)$ in L^2 .

Proof: Since $S_m(x) \rightarrow S(x)$ uniformly,

$\forall \varepsilon > 0, \exists N = N(\varepsilon) \in \mathbb{N}$ s.t.

$m \geq N$ implies $\max_{x \in [a, b]} |S_m(x) - S(x)| < \sqrt{\frac{\varepsilon}{b-a}} \quad (*)$

Then $m \geq N$ implies

$$\begin{aligned} \|S_m - S\|_{L^2}^2 &= \int_a^b |S_m(x) - S(x)|^2 dx \\ &\stackrel{\text{by } (*)}{\leq} \int_a^b \left(\sqrt{\frac{\varepsilon}{b-a}} \right)^2 dx = \varepsilon \end{aligned}$$

$\therefore S_m(x) \rightarrow S(x)$ in L^2 .

(2) If $S_m(x) \rightarrow S(x)$ uniformly,
then show $S_m(x) \rightarrow S(x)$ pointwise.

Proof: By def'n, since $S_m(x) \rightarrow S(x)$ uniformly,
we have $\forall \varepsilon > 0, \exists N = N(\varepsilon)$ s.t.

$$n \geq N \text{ implies } \max_{a \leq x \leq b} |S_m(x) - S(x)| < \varepsilon.$$

Pick this $N = N(\varepsilon)$ in the ε - N game for
pointwise convergence. We are done, since
 $\forall x \in [a, b], |S_m(x) - S(x)| \leq \max_{a \leq x \leq b} |S_m(x) - S(x)| < \varepsilon$

S.4 #3 $\gamma_n \rightarrow \infty$

$$f_n(x) \stackrel{\text{def}}{=} \begin{cases} \gamma_n & \text{if } x \in [\frac{1}{2} - \frac{1}{n}, \frac{1}{2}) \\ 0 & \text{if } x = \frac{1}{2} \\ -\gamma_n & \text{if } x \in (\frac{1}{2}, \frac{1}{2} + \frac{1}{n}] \\ 0 & \text{otherwise} \end{cases}$$

Show: (a) $f_n(x) \rightarrow 0$ pointwise

If $x = \frac{1}{2}$ or $|x - \frac{1}{2}| \geq 1$, then $f_n(x) \equiv 0$
Otherwise we have $0 < |x - \frac{1}{2}| < 1$.

We may choose N large enough s.t. $\frac{1}{N} > |x - \frac{1}{2}|$,
so that for $n \geq N$, $f_n(x) \equiv 0$.

Thus $f_n(x) \rightarrow 0$ pointwise

(b) If $f_n(x) \rightarrow 0$ uniformly, then
 $\max_{x \in \mathbb{R}} |f_n(x) - 0| \rightarrow 0$ as $n \rightarrow \infty$.

But on the other hand, by def'n of $f_n(x)$
and that $\gamma_n \rightarrow \infty$ as $n \rightarrow \infty$.

$$\max_{x \in \mathbb{R}} |f_n(x) - 0| = |\gamma_n| \rightarrow \infty \text{ as } n \rightarrow \infty$$

(c)
$$\int_{-\infty}^{\infty} |f_n(x) - 0|^2 dx$$

$$= \int_{\frac{1}{2} - \frac{1}{n}}^{\frac{1}{2}} |f_n(x) - 0|^2 dx + \int_{\frac{1}{2}}^{\frac{1}{2} + \frac{1}{n}} |f_n(x) - 0|^2 dx$$

(the other parts vanish)

$$= \int_{\frac{1}{2} - \frac{1}{n}}^{\frac{1}{2}} |\gamma_n(x)|^2 dx + \int_{\frac{1}{2}}^{\frac{1}{2} + \frac{1}{n}} |\gamma_n(x)|^2 dx$$
$$= \int_{\frac{1}{2} - \frac{1}{n}}^{\frac{1}{2}} n^{\frac{2}{3}} dx + \int_{\frac{1}{2}}^{\frac{1}{2} + \frac{1}{n}} n^{\frac{2}{3}} dx$$
$$= n^{\frac{2}{3}} \cdot \frac{1}{n} + n^{\frac{2}{3}} \cdot \frac{1}{n} = 2n^{-\frac{1}{3}} \rightarrow 0$$

as $n \rightarrow \infty$

Thus $f_n(x) \rightarrow 0$ in L^2 if $\gamma_n = n^{1/3}$.

(d)
We will just show $f_n(x) \not\rightarrow 0$ in L^2 if $\gamma_n = n$.

(In fact, if $f_n \rightarrow f$ in L^2 for some f and $f_n \rightarrow g$ pointwise for some g , then $f = g$ "almost everywhere".
The "fact" is non-trivial, using measure theory)

$$\begin{aligned} & \int_{-\infty}^{\infty} |f_n(x) - 0|^2 dx \\ &= \int_{\frac{1}{2} - \frac{1}{n}}^{\frac{1}{2}} |f_n(x) - 0|^2 dx + \int_{\frac{1}{2}}^{\frac{1}{2} + \frac{1}{n}} |f_n(x) - 0|^2 dx \\ & \quad \text{(the other parts vanish)} \\ &= \int_{\frac{1}{2} - \frac{1}{n}}^{\frac{1}{2}} |\gamma_n(x)|^2 dx + \int_{\frac{1}{2}}^{\frac{1}{2} + \frac{1}{n}} |\gamma_n(x)|^2 dx \\ &= \int_{\frac{1}{2} - \frac{1}{n}}^{\frac{1}{2}} n^2 dx + \int_{\frac{1}{2}}^{\frac{1}{2} + \frac{1}{n}} n^2 dx \end{aligned}$$

$$= 2n \rightarrow +\infty \text{ as } n \rightarrow \infty.$$

Thus $f_n(x) \not\rightarrow 0$ in L^2 if $\gamma_n = n$.